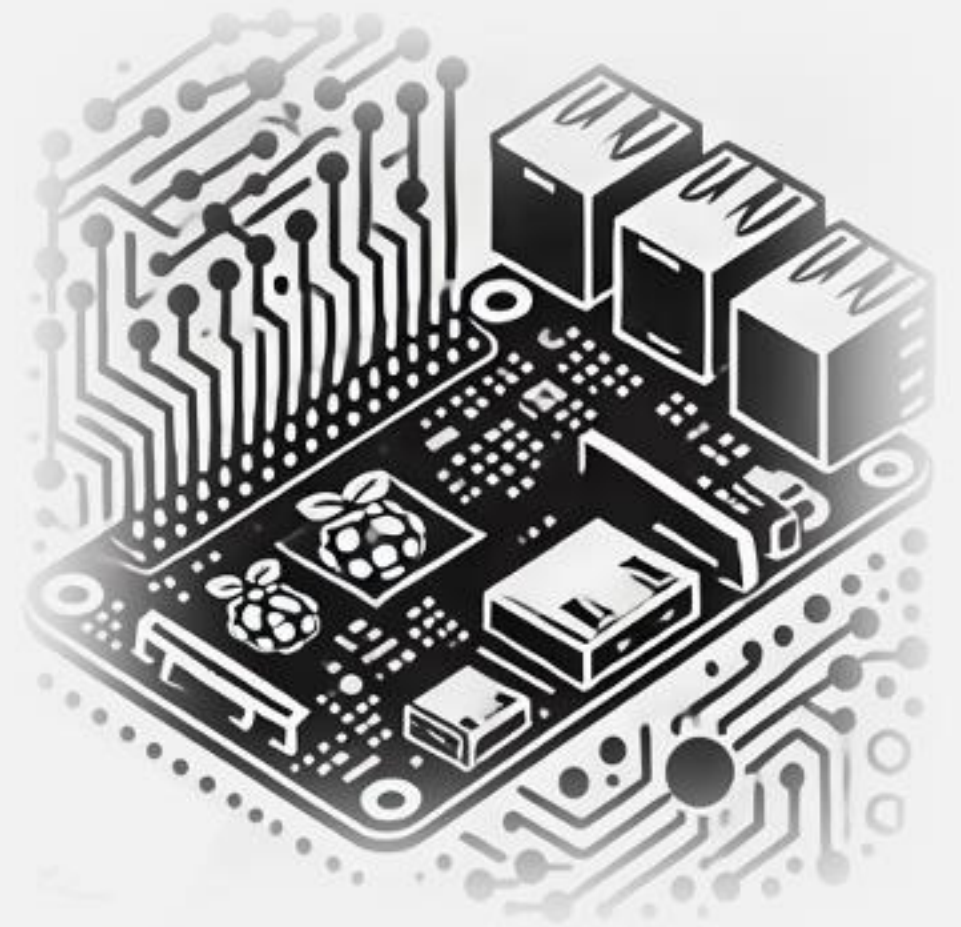


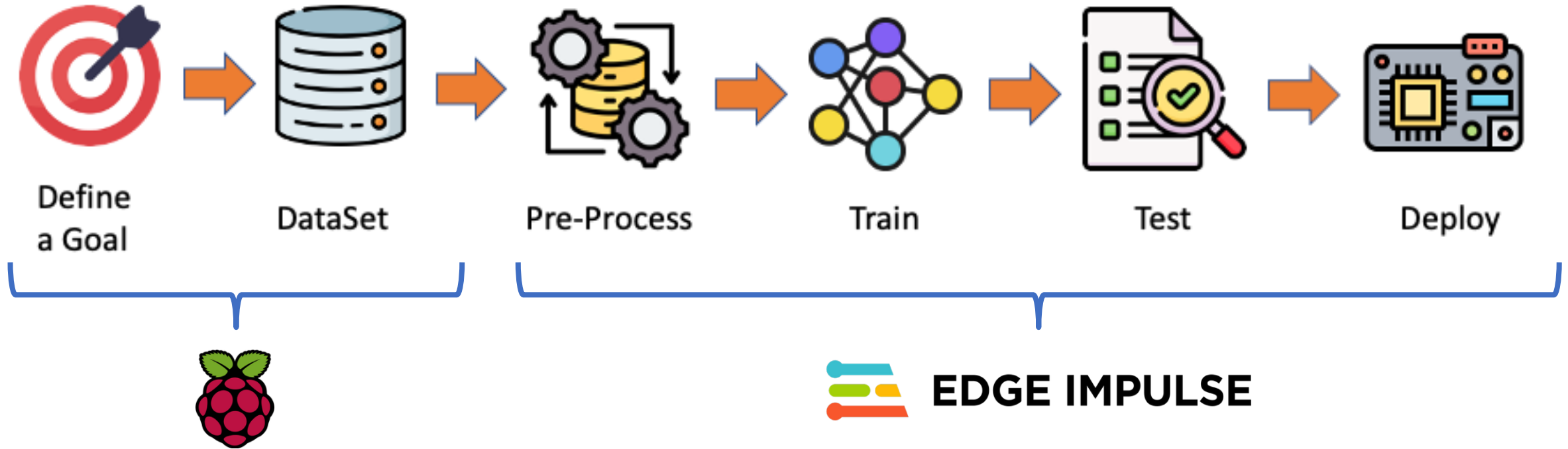
IESTI05 – Edge AI

Machine Learning System Engineering

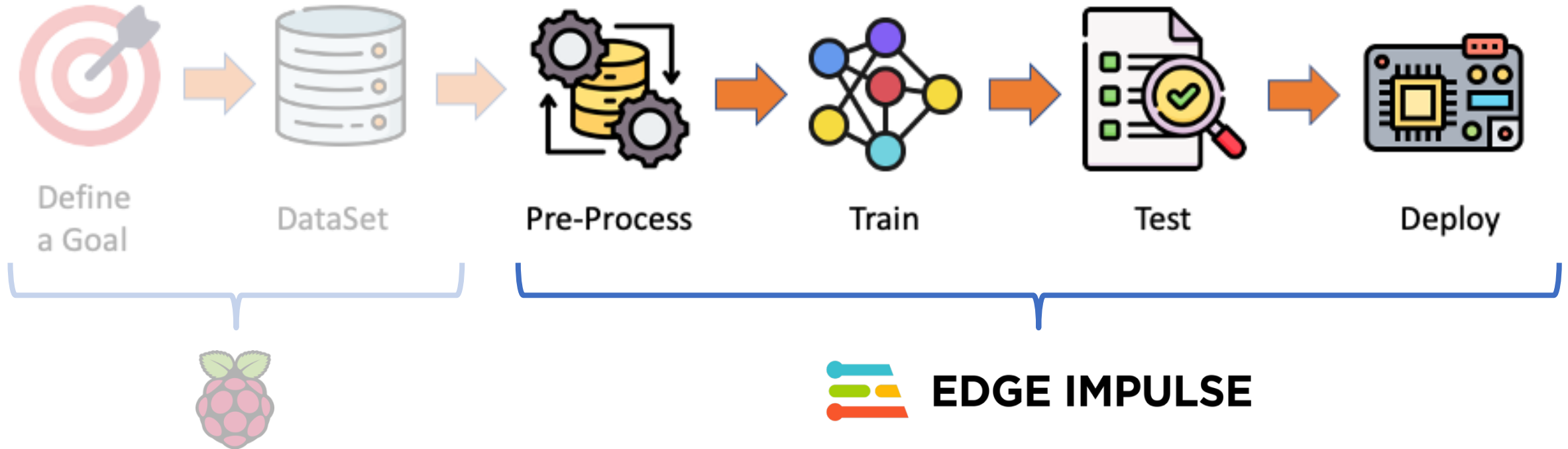
7. Image Classification Project: Training, Test & Deploy at EI



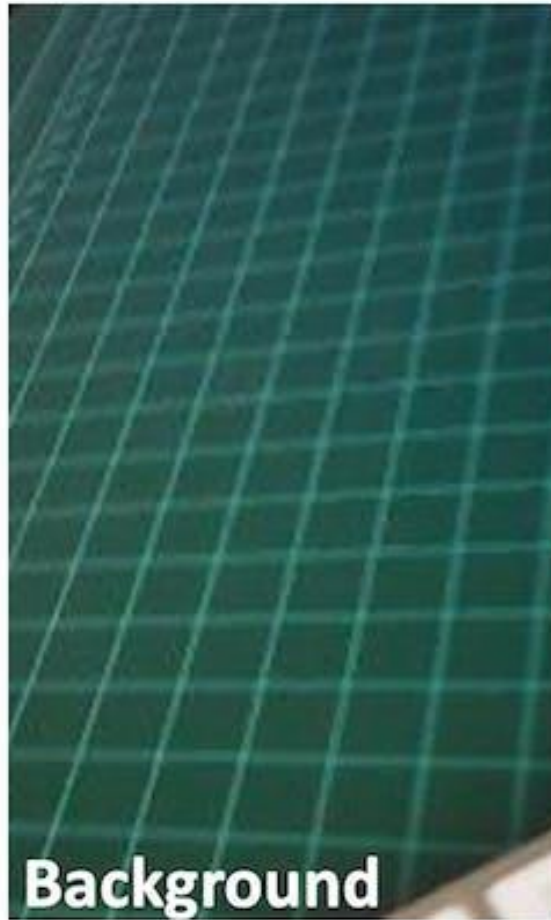
Machine Learning Workflow



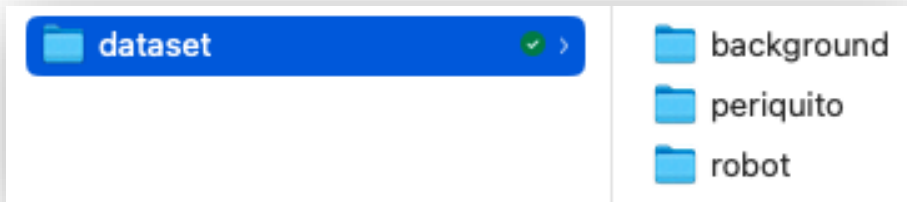
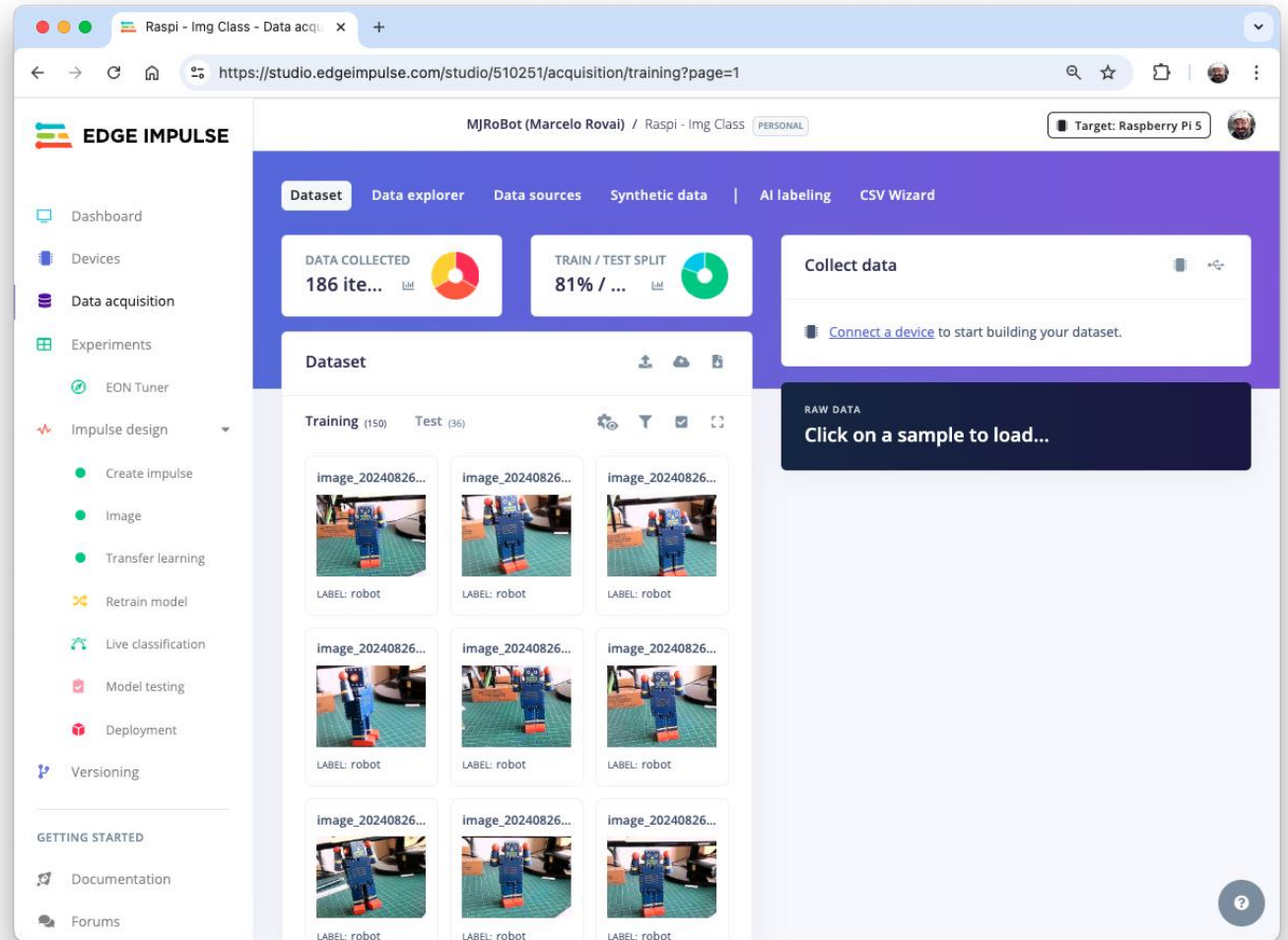
Machine Learning Workflow



The Goal



Data Aquisition



Pre-Processing



Image data

Input axes
image

Image wi...
160

Image he...
160

Resize mode
Sql

Image

Name
Image

Input axes (1)
☒ Image

EDGE IMPULSE

Dashboard

Devices

Data acquisition

Experiments

EON Tuner

Impulse design

Create impulse

Image

Transfer learning

Retrain model

Live classification

Model testing

Deployment

Versioning

GETTING STARTED

Documentation

Raspi - Img Class - Image - E

https://studio.edgeimpulse.com/studio/510251/impulse/1/dsp/image/3

MJRoBot (Marcelo Rovai) / Raspi - Img Class

Target: Raspberry Pi 5

Parameters

Generate features

Raw data

Show: All labels

image_20240826-201348 (r1



Raw features

0x372b2f, 0x362a2e, 0x504448, 0x665a5e, 0x7d7175, 0x918589, ...

Parameters

Image

Color depth ③

RGB

Save parameters

DSP result

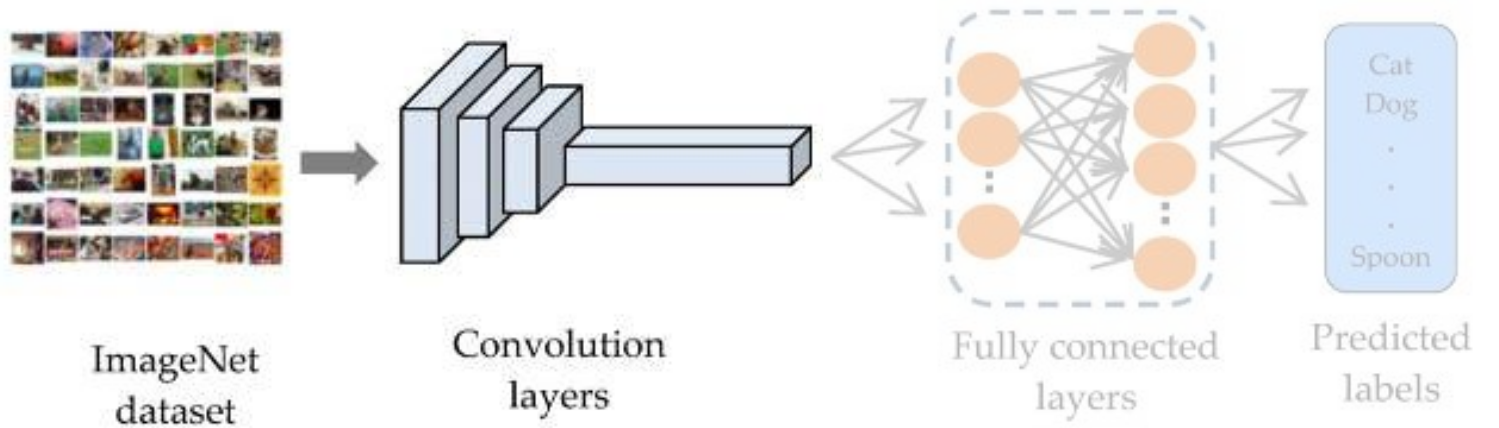
Image



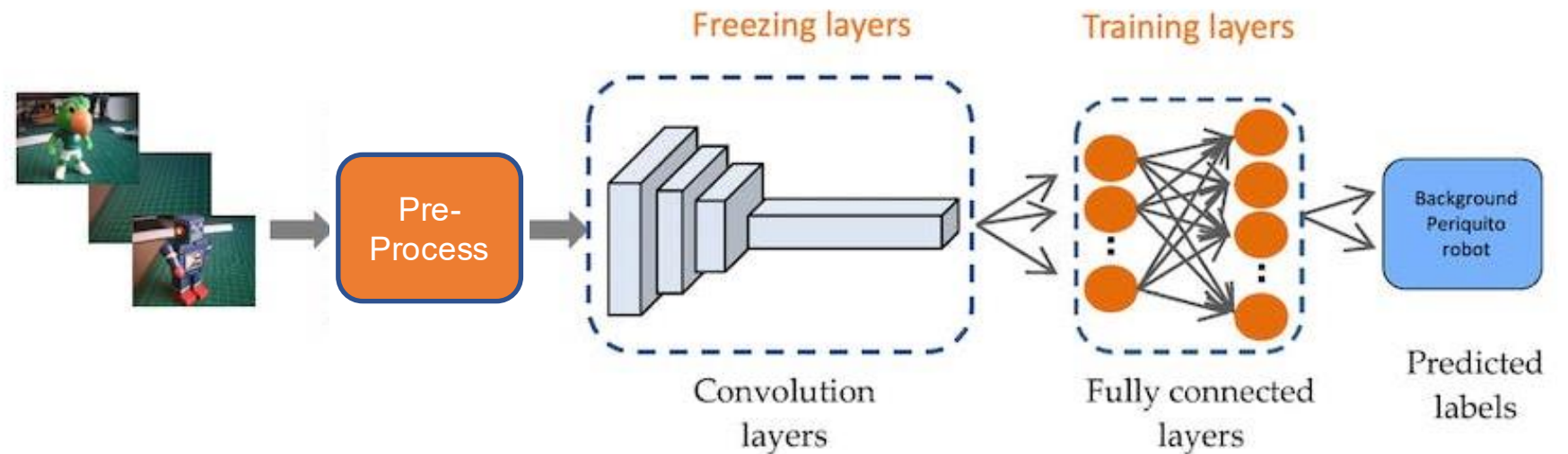
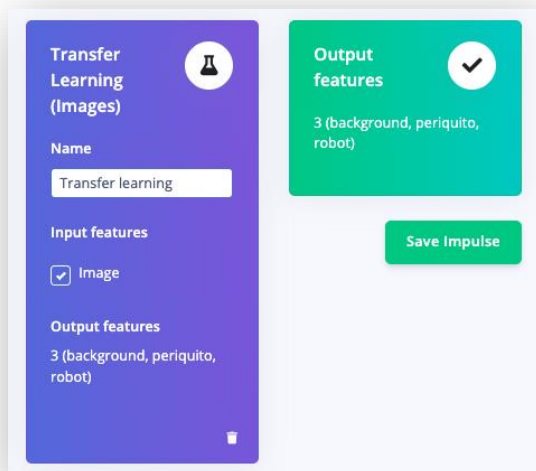
Processed features

0.2157, 0.1686, 0.1843, 0.2118, 0.1647, 0.1804, 0.3137, 0.26

Model



MobileNetV2 160x160 1.0



Train & Test



Neural Network settings

Training settings

Number of training cycles ⓘ

20

Use learned optimizer ⓘ

☐

Learning rate ⓘ

0.0005

Training processor ⓘ

CPU

Data augmentation ⓘ

☒

Advanced training settings

Validation set size ⓘ

20

%

Split train/validation set on metadata key ⓘ

Batch size ⓘ

32

Auto-weight classes ⓘ

☐

Profile int8 model ⓘ

☒

Neural network architecture

Input layer (76,800 features)

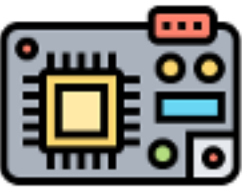
 MobileNetV2 160x160 1.0 (no final dense layer, 0.1 dropout)

Choose a different model

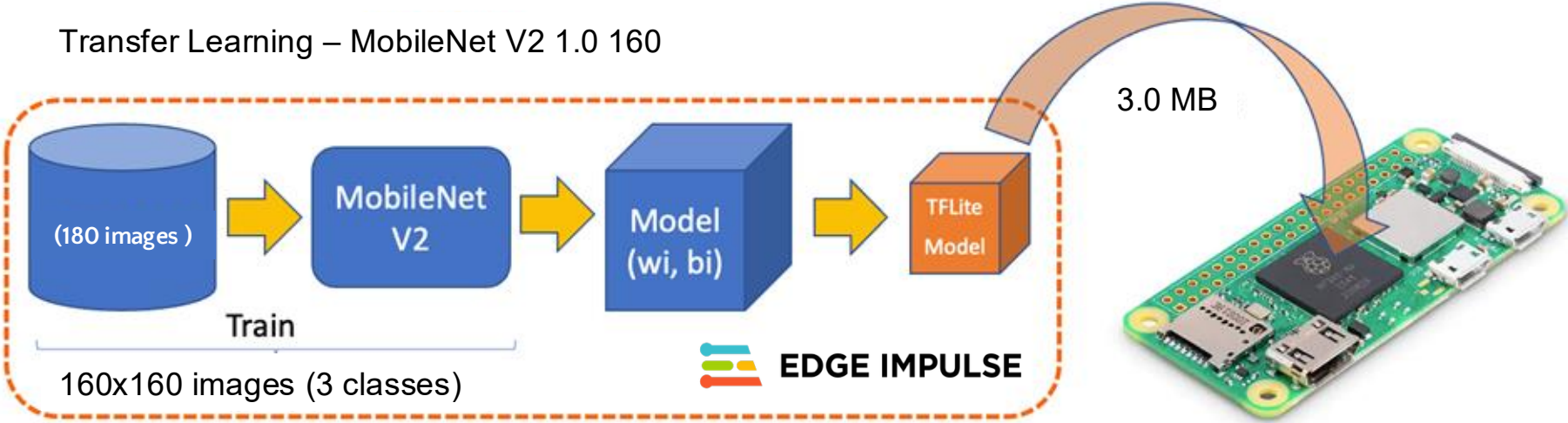
Output layer (3 classes)



Deploy



Transfer Learning – MobileNet V2 1.0 160



Download block output			
TITLE	TYPE	SIZE	
Image training data	NPY file	150 windows	
Image training labels	NPY file	150 windows	
Image testing data	NPY file	36 windows	
Image testing labels	NPY file	36 windows	
Transfer learning model	TensorFlow Lite (float32)	9 MB	
Transfer learning model	TensorFlow Lite (int8 quantized)	3 MB	
Transfer learning model	Model evaluation metrics (JSON file)	5 KB	
Transfer learning model	TensorFlow SavedModel	8 MB	
Transfer learning model	Keras h5 model	8 MB	

[Edge Impulse Project: Raspi - Img Class](#)



Raspberry Pi Inference:

[30_Image_Classification_edge_impulse.ipynb](#)



Live Image Classification

[img_class_live_infer.py](#)

```
python img_class_live_infer.py
```

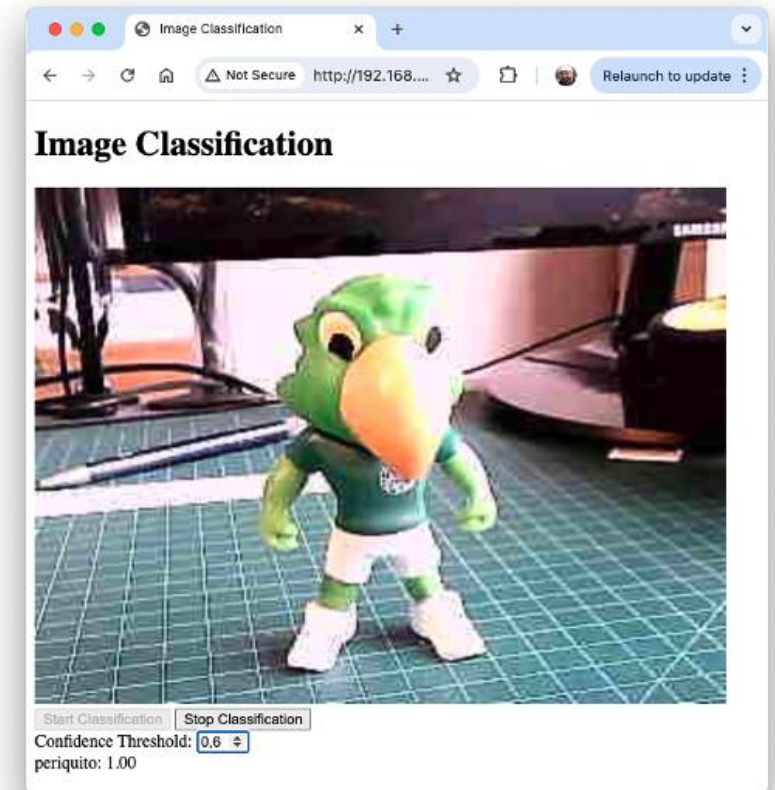
* Running on <http://192.168.4.210:5000>
Press CTRL+C to quit

```
marcelo_rovai — mjrovai@raspi-zero2w-26: ~/Documents/TFLITE/IMG_CLASS — ssh mjrovai@192.168.4.210 — 92...
GNU nano 8.4 img_class_live_infer.py
from flask import Flask, Response, render_template_string, request, jsonify
from picamera2 import Picamera2
import io
import threading
import time
import numpy as np
from PIL import Image
from ai_edge_litert.interpreter import Interpreter
from queue import Queue

app = Flask(__name__)

# Global variables
picam2 = None
frame = None
frame_lock = threading.Lock()
is_classifying = False
confidence_threshold = 0.8
model_path = "./models/ei-raspi-img-class-int8-quantized-model.tflite"
labels = ['background', 'periquito', 'robot']
interpreter = None
classification_queue = Queue(maxsize=1)

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line
```



Questions?



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